

TEAM OUTLIER

Data Analysis & Visualisation

Healthtrace Hackathon Documentation

Turning Fragmented Data into Actionable Public-Health Signals

Detecting likely cases from vitals and mapping exposure across the Tracy devices deployed

1. Problem Statement

Given three datasets from a Tracy field deployment, our team chose to answer one operational question: does this telemetry actually carry a defensible early-warning signal a public-health officer can act on, and if so, where does it plug into the existing surveillance architecture?

2. Dataset Description

We worked with three files. Each one is a separate lens on the same problem.

File	Records	Window	Key columns used
vitals.csv	2,307	13 May – 3 Jun 2024	device_id (D001–D021), temperature, heartbeat, movement, battery, geohash, lat/lon
mobility.csv	50,737	29 Jan – 22 Mar 2024	user_id (U001–U079), geohash, has_contact, exposure_score, daily_exposure, close_contacts
contact_tracing.csv	119,338	19 Oct – 18 Dec 2023	user_id, geohash, mac (proximate device), rssi, proximity (close / very close / moderate / distant / unscored)

Two structural facts shaped our analysis:

- **Datasets do not overlap in time** — vitals are from May–June 2024, mobility from January–March 2024, contacts from October–December 2023, so they cannot be row-joined.
- **Identifier spaces differ** — vitals device IDs (D001–D021) are not the same identifiers used in the contact-tracing MAC field (D01000–D22000 range, distinct devices entirely), so the contact graph cannot be linked to a specific wearer's vitals.

We therefore approached the three files as complementary lenses on the same area, joined at the spatial-cell × day grid rather than row by row.

3. Methodology

Our team worked in a single Jupyter notebook (Python 3, pandas, scipy, scikit-learn, networkx, matplotlib) in three deliberate stages.

3.1 Inspect before cleaning

Before any filtering, we built per-channel distributions and a per-device reliability table. We deliberately did not drop the suspicious 42 °C readings as outliers — those readings are themselves a finding. We computed a per-device run-length feature (count of consecutive identical temperature values) so any later step could distinguish a frozen sensor from a genuinely stable reading.

3.2 Audit the hardware

We declared a device faulty under any of four rules: (i) a stuck-temperature run of 50 or more identical samples; (ii) a battery reading of 0 in at least 50 % of the device's records; (iii) a maximum temperature ≥ 42 °C (incompatible with a living wearer, since core temperatures above ~ 41 °C cause organ failure); or (iv) a temperature standard deviation ≤ 0.5 °C across all the device's samples. A single rule was sufficient — we did not require multiple agreements.

3.3 Extract signal from cleaned data

We then ran the four required visualisations — temperature distribution, heart-rate distribution, device activity over time, anomaly timeline — and tested the central biological hypothesis: in a real febrile population, heart rate should rise by 8–10 bpm per °C above 37 °C (Liebermeister's rule). We computed Pearson and Spearman correlations between temperature and heart rate after removing stuck-sensor runs.

4. Key Findings

Finding 1 — The temperature channel is systematically miscalibrated

The modal reading across all 2,307 samples is 34 °C — roughly 2.1 to 3.2 °C below the healthy adult core range (36.1–37.2 °C). The likely cause is that the wearable measures skin or wristband temperature rather than core temperature. A raw absolute-temperature threshold therefore cannot be used as a febrile screen. The device needs either hardware calibration to a clinical reference or, more pragmatically, a per-device baseline scheme where alerts fire on deltas from each device's own running median.

Figure 1 — Temperature distribution across all 2,307 readings. The mode sits at 34 °C, well below the healthy adult core band; the dense bar at exactly 42 °C is biologically impossible and traces to two specific broken devices.

Finding 2 — Heart rate is plausible but does not track temperature

Heart rate looks reliable: most readings sit around 70 bpm, which is normal. But when we tested whether heart rate rises with fever (as Liebermeister's rule predicts), the correlation was essentially zero. The dataset does not yet show a clear biological outbreak signal.

Finding 3 — A single broken device produced 60 % of the data

Device D012 alone contributed 1,375 of the 2,307 vitals readings (59.6 %), while its battery field read exactly 0 across every record (inconsistent with the device still transmitting), and its temperature value was stuck at exactly 34 °C for a run of 979 consecutive samples. Device D020 produced a separate 108-reading run at exactly 42 °C — a textbook frozen-sensor signature. Across the fleet, nine of the twenty-one devices flagged faulty on at least one rule; those nine devices contributed 92 % of all readings.

Figure 2 — Left: readings per device on a log scale, with the nine flagged-faulty devices in red. Right: D020's full temperature stream — the device behaved normally then snapped to a 42 °C plateau.

Finding 4 — Hardware anomalies vastly outnumber biological anomalies

Looking at daily alerts, hardware problems (dead batteries, stuck sensors) far outnumber genuine biological signals (fever plus elevated heart rate). If both feeds share a single dashboard, the officer will mostly see false alarms from broken devices, learn to ignore the system, and a genuine outbreak signal could be missed.

Finding 5 — Best spatial resolution is ~5 km grids

- Large grids (~20 km) are too coarse — they lump whole districts together, so teams cannot tell where to go.
- Very small grids (~1–2 km) are too fine — they generate too many meaningless alerts.
- Medium grids (~5 km) work best — precise enough to surface real hotspots, broad enough to absorb GPS noise.

5. Limitations & Assumptions

- **No ground truth:** no confirmed cases or lab results, so all signals are inferred.
- **No temporal overlap:** datasets cover different quarters; confirmation is by area, not by individual.
- **Small fleet:** only 21 devices over 22 days, dominated by one faulty sensor — feasibility scale, not deployment scale.
- **Separate ID spaces:** vitals, contact-tracing, and mobility IDs do not align, so individuals cannot be tracked across all streams.
- **Device mapping assumed:** each device treated as one wearer; if devices were shared, statistics may mix multiple people.

6. Recommendations for Public-Health Officers

- **Separate dashboards:** route hardware faults (battery, stuck sensors) to field ops, and biological signals (fever, exposure) to clinical teams. Mixing them causes alert fatigue.
- **Per-device baselines:** flag fevers by changes from each device's own median, not a universal 38 °C cutoff. This avoids miscalibration errors.
- **Integrate with SORMAS:** add a wearable-proximity module that feeds exposure scores and 5 km hotspot maps directly into Nigeria's existing surveillance system.
- **Default to geohash-5:** use 5 km grids for alerts — precise enough for neighbourhood targeting, broad enough to absorb GPS noise. Drill down only when a hotspot is flagged.

7. Future Work — Four Next Steps

- **Live data pipeline:** build a feed that summarises daily exposure and hotspot data in a format SORMAS can consume directly.
- **Supervised model:** once lab-confirmed cases exist, train the system to learn risk-score weights from real outcomes.
- **Adaptive baselines:** let each device set its own temperature baseline, improving accuracy without waiting for new hardware.
- **CHW alert app:** create a simple mobile tool showing the top-10 priority visits, with one-tap outcome capture to close the feedback loop.